

02LQD Specification and Simulation of Digital Systems

2023/04/13

1. Combinational circuit design (2 points)

Thermometer-to-binary (TM2B) encoders are important components of Flash Analog-to-Digital Converters (FADC). The thermometer outputs a pure binary signal X on 7 bits. The encoder receives X and generates Z on 3 bits according to the following table:

X	Z
0000000	000
0000001	001
0000011	010
0000111	011
0001111	100
0011111	101
0111111	110
1111111	111

Design in VHDL an encoder implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSEIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (12 points)

A Mealy machine acts as a recognizer of sequences transmitted on a serial line. It has 2 inputs and 2 outputs:

- the 1-bit input signal En enables machine operation when it is 1. Whenever it is 0 the machine returns to its initial state and waits for a new sequence
- the 1-bit input signal X is the serial line
- the 2-bit output signal $Z = Z_1Z_0$ is 10 when the input sequence 111 has been recognized on the last 3 bits, 01 when the input sequence 00 has been recognized on the last 2 bits, 00 otherwise.

Reset is synchronous.

Example

En	00111111111111110
X	01010001101001111
Z_1	000000000000000110
Z_0	000001100000100000

Design a **Mealy** FSM with the above specified inputs, outputs and behavior:

- Draw the state diagram (4 points)
- Design the FSM resorting to the VHDL behavioural description style with **1 process** with the same timing behaviour as the 2-process description (4 points).
- Design the FSM resorting in Verilog with **2 always blocks** (4 points).

3. FSM-D design (16 points)

Multiplication is a complex operation. A non-optimized algorithm is based on repetitive addition. For example $-7 \times 5 = -(7 + 7 + 7 + 7 + 7)$. Design a FSM-D to perform multiplication of signed binary numbers by repeated addition. Inputs a and b are signed binary numbers on 8 bits. Define a proper size for the output z . Computation starts when a `start` signal is asserted for one clock cycle. When computation is over the `ready` signal is asserted for one clock cycle. Reset is asynchronous. Values for output z are available at each computation step, but they make sense only when `ready` is 1.

Steps:

1. Define a proper size for the output z (**2 points**)
2. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**6 points**)
3. Design a FSM-D:
 - create 2 processes for behavioural datapath description from the HLSM state diagram (**4 points**)
 - create 2 processes for behavioural controller description from the HLSM state diagram. (**4 points**).

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
 - running VHDL code and testbench
 - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

no later than Sunday April 16 11.59 pm

- format of .zip archive name surname_ID (example Rossi_244123)
- remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.